Description

Image





Caption

1. Bronze bull sculpture, New York. 2. Bronze door knocker. Images used under license from Shutterstock.com

The material

Bronze, the material of the Bronze age (3000 - 1000 BC) was originally an alloy of copper (Cu) and tin (Sn), but today the term is used for any alloy of copper that has a principal alloying element other than zinc or nickel. The tin bronzes contain 5 - 25% Sn; those with more than 10% Sn have high strength but little ductility, so they must be cast to shape. Aluminum bronzes (Cu with 4 - 14% Al plus a little Mn) have high strength and excellent corrosion resistance. Phosphor bronzes (Cu with 1.25 - 10% Sn and up to 0.6% P) have low friction coefficient and are widely used for bearings. Gunmetal is a tin bronze containing some zinc and lead.

Compositional summary

Tin bronze: Cu + 5 - 25% Sn
Aluminum bronze: Cu + 4 - 14% Al
Phosphor bronze: Cu + 3 - 9% Sn + 0.1 - 0.4% P
Gunmetal: Cu + 5 % Sn + 5 % Zn + 5 % Pb

General properties

Density		8.05e3	-	8.7e3	kg/m^3
Price	*	7.65	-	8.04	GBP/kg
Date first used		-3000			

Mechanical properties

Young's modulus		97	-	130	GPa
Shear modulus	*	38	-	49	GPa
Bulk modulus	*	120	-	140	GPa
Poisson's ratio		0.34	-	0.35	
Yield strength (elastic limit)		130	-	509	MPa

Tensile strength		322	-	700	MPa
Compressive strength	*	130	-	509	MPa
Elongation		5.3	-	55	% strain
Hardness - Vickers		75.6	-	200	HV
Fatigue strength at 10^7 cycles	*	154	-	254	MPa
Fracture toughness	*	28.8	-	67.9	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	2.8e-5	-	1.6e-4	

Thermal properties

Melting point		947	-	1.08e3	°C
Maximum service temperature		180	-	255	°C
Minimum service temperature		-273			°C
Thermal conductor or insulator?		Good conductor			
Thermal conductivity		38	-	62	W/m.°C
Specific heat capacity		380	-	398	J/kg.°C
Thermal expansion coefficient		15.5	-	18.2	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good conductor			
Electrical resistivity		14	-	21	µohm.cm

Optical properties

Transparency		Opaque			

Critical Materials Risk

High critical material risk?		Yes			

Processability

Castability		4	-	5	
Formability		3	-	5	
Machinability		4	-	5	
Weldability		4	-	5	
Solder/brazability		4	-	5	

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			
Soils, acidic (peat)		Excellent			
Soils, alkaline (clay)		Excellent			
Wine		Excellent			

Durability: acids

Acetic acid (10%)	Limited use
Acetic acid (glacial)	Unacceptable
Citric acid (10%)	Acceptable
Hydrochloric acid (10%)	Limited use
Hydrochloric acid (36%)	Unacceptable
Hydrofluoric acid (40%)	Unacceptable
Nitric acid (10%)	Acceptable
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Limited use
Phosphoric acid (85%)	Unacceptable
Sulfuric acid (10%)	Limited use
Sulfuric acid (70%)	Unacceptable

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Excellent
Crude oil	Acceptable
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Acceptable
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Excellent
Acetone	Excellent
Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	

	Excellent
Formaldehyde (40%)	Excellent
Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Excellent
Fluorine (gas)	Acceptable
O2 (oxygen gas)	Acceptable
Sulfur dioxide (gas)	Excellent

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Excellent
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Acceptable
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	1.85e7	tonne/yr
Reserves, principal component	6.9e8	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 5.2	-	5.73	kg/kg
Embodied energy (virgin grade production)	* 77.3	-	85.2	MJ/kg
Water usage	* 799	-	883	l/kg

Material processing: energy

Metal casting, embodied energy	* 40.8	-	47	MJ/kg
Metal extrusion, hot, embodied energy	* 23.2	-	26	MJ/kg

Metal rolling and forging, embodied energy	* 12.1	-	13.6	MJ/kg
Wire drawing, embodied energy	* 13.1	-	14.5	MJ/kg
Metal powder forming, embodied energy	* 36.9	-	46.1	MJ/kg
Chemical vapor deposition (CVD), embodied energy	* 8.15e3	-	9.83e3	MJ/kg
Machining, coarse, embodied energy	* 18	-	19.9	MJ/kg
Machining, fine, embodied energy	* 24.7	-	27.4	MJ/kg
Machining, grinding, embodied energy	* 31.4	-	34.9	MJ/kg
Machining, laser, CO2, embodied energy	* 0.97	-	1.17	MJ/m

Material processing: climate change

Metal casting, climate change (CO2-eq)	* 2.86	-	3.33	kg/kg
Metal extrusion, hot, climate change (CO2-eq)	* 1.52	-	1.7	kg/kg
Metal rolling and forging, climate change (CO2-eq)	* 0.793	-	0.889	kg/kg
Wire drawing, climate change (CO2-eq)	* 0.86	-	0.95	kg/kg
Metal powder forming, climate change (CO2-eq)	* 2.99	-	3.74	kg/kg
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 671	-	808	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.17	-	1.3	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.61	-	1.79	kg/kg
Machining, grinding, climate change (CO2-eq)	* 2.05	-	2.29	kg/kg
Machining, laser, CO2, climate change (CO2-eq)	* 0.0799	-	0.0962	kg/m

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 1.14	-	1.26	kg/kg
Embodied energy, recycling	* 18.6	-	20.4	MJ/kg
Recycle fraction in current supply	28			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Bronzes are readily recycled.

Supporting information

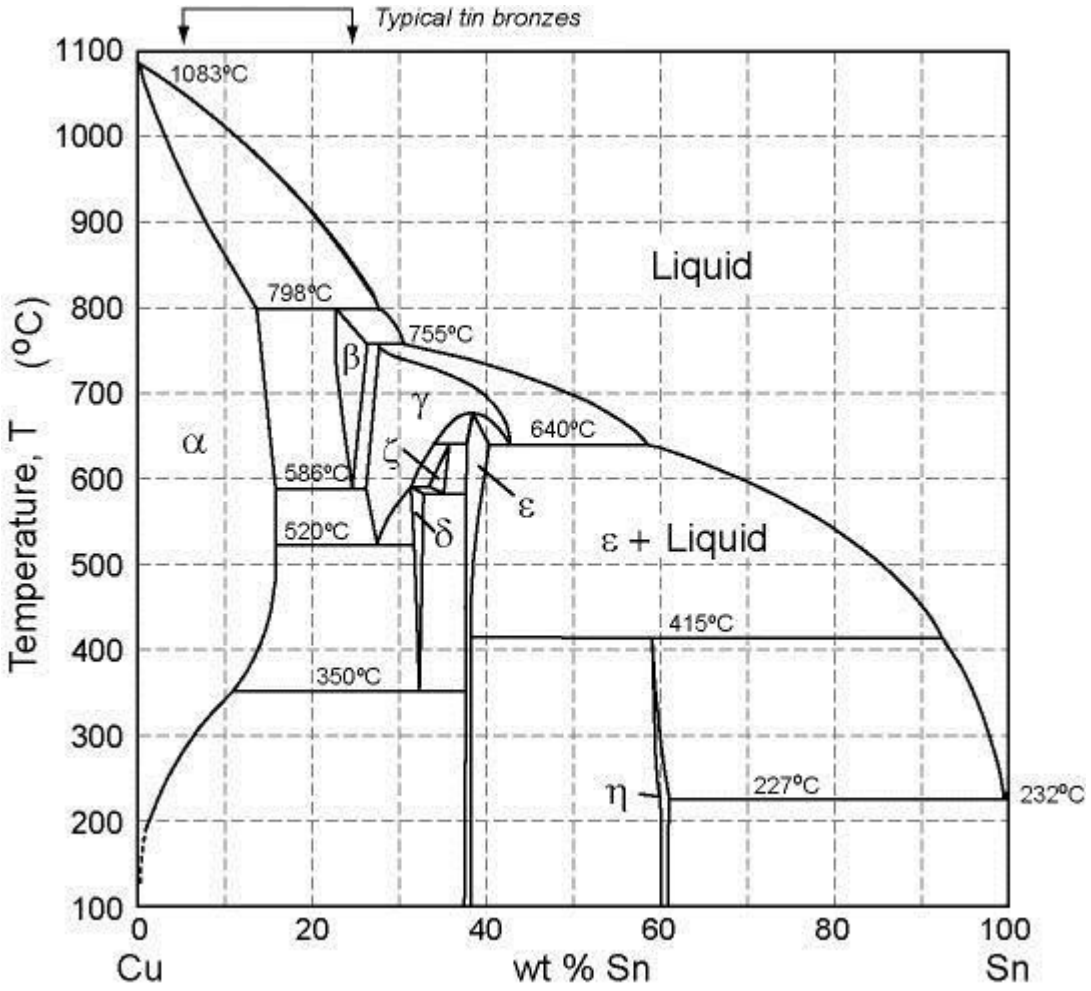
Design guidelines

Bronze, the material of the Bronze age (3000 - 1000 BC) is still the material of choice for grand statuary, bells and ornamental architectural parts. It is hard and strong (hence its use for weapons), it can be cast to intricate shapes and it resists corrosion well even in a marine environment. Low-tin and aluminum bronzes can be rolled to sheet and worked easily; those with higher alloy content must be cast.

Technical notes

There is now a UNS designation system for copper and its alloys: the letter C (for 'copper') followed by a 5-digit number. Only the first digit means anything: C1**** designates almost pure copper, the C2, C3 and C4 series are brasses with increasing zinc content, the C5s are bronzes based on copper and tin, the C6s are other bronzes containing aluminum instead of tin, and the C7s are copper-nickel alloys.

Phase diagram



Phase diagram description

Tin bronzes are alloys of copper (Cu) with 5 - 25% tin (Sn), for which this is the phase diagram.

Typical uses

Tin bronzes and gunmetals: sand casting of marine and architectural fittings, bells, pump parts, taps and valves, coinage. Aluminum bronzes: die-cast and wrought components, particularly those to resist corrosion, heat exchangers, condensers, ships propellers and marine fittings, architectural cladding, valve and pump bodies. Phosphor bronzes: bearings and gears, springs, pump parts, coinage, tubing a plate to resist corrosion and erosion.

Tradenames

Coinage bronze (Cu 3% Sn 1.5% Zn)

Links

ProcessUniverse

Producers

Reference